

1) a) On excel file ↓

b) $2Rr = 120 \text{ min}$

$$C_A(t) = C_{A\infty} + (C_{A0} - C_{A\infty})e^{-kt}$$

$$C_A(120) = 0.0495 + (0.1823 - 0.0495)e^{-0.0093 \times 120} = 0.0931 \text{ g/L}$$

$$\Delta C = C_{A0} - C_A(120)$$

$$= 0.1823 - 0.0931 = 0.0892 \text{ g/L}$$

$$30.5 \text{ gal} \times 3.785 = 115.46 \text{ L}$$

$$\Rightarrow m_0 = \Delta C \times V = 0.0892 \frac{\text{g}}{\text{L}} \times 115.46 \text{ L} = 10.3 \text{ g}$$

2) Excel file ↓

4) Calculate Hydrocarbon Emissions in 1970:

$$14 \frac{\text{g}}{\text{mile}} \times 1.1 \times 10^{12} \text{ mile} = 1.54 \times 10^{13} \text{ g}$$

$$\frac{1.54 \times 10^{13}}{453.6} \approx 3.39 \times 10^{10} \text{ lb}$$

$$\frac{3.39 \times 10^{10}}{2000} \approx 1.70 \times 10^7 \text{ tons}$$

At 1970 $\approx$ 17 million tons

Total Hydrocarbon Emissions in 2000

$$1 \frac{\text{g}}{\text{mile}} \times 2.6 \times 10^{12} \text{ mile} = 2.6 \times 10^{12} \text{ g}$$

$$\frac{2.6 \times 10^{12}}{453.6} = 5.73 \times 10^9 \text{ lb}$$

$$\frac{5.73 \times 10^9}{2000} = 2.9 \times 10^6 \text{ tons}$$

Total emissions in 2000 $\approx$ 2.9×10^6 tons

Percent change (1970 - 2000)

$$\% \Delta = \frac{2.9 - 1.7}{1.7} \times 100\% = -83\%$$

✓ 15/15

⇒ The total hydrocarbon emissions decreased from approximately 17 million tons in 1970 to 2.9 million tons in 2000, representing an approximate 83% decrease over 30 years period, despite a significant increase in total vehicle miles traveled

5)

$$\begin{aligned} \rho_{\text{water}} &\approx 0.9982 \text{ g/mL} \\ \rho_{\text{ethanol}} &\approx 0.7893 \text{ g/mL} \\ M_{\text{ethanol}} &= 46.07 \text{ g/mol} \end{aligned}$$

We have: 60wt% water ⇒ 40wt% ethanol

a)

$$\begin{aligned} m_w &= 60\text{g} \\ m_e &= 40\text{g} \end{aligned}$$

$$V = \frac{m_w}{\rho_w} + \frac{m_e}{\rho_e} = \frac{60}{0.9982} + \frac{40}{0.7893} = 110.79 \text{ mL}$$

$$\rho_{\text{mix}} = \frac{100\text{g}}{110.79 \text{ mL}} = 0.90264 \text{ g/mL}$$

⇒ Specific gravity = 0.903

$$m_e = 150 \text{ mol} \times 46.07 \text{ g/mol} = 6910.27 \text{ g}$$

$$m_{\text{mix}} = \frac{m_e}{0.40} = \frac{6910.27}{0.4} = 17275.67 \text{ g}$$

$$V_{\text{mix}} = \frac{m_{\text{mix}}}{\rho_{\text{mix}}} = \frac{17275.67}{0.90264} = 19139 \text{ mL} = 19.139 \text{ L}$$

b) SG = 0.93518 ⇒ $\rho_{\text{mix}} = 0.93518 \text{ g/mL}$

$$V_{\text{true}} = \frac{m_{\text{mix}}}{\rho_{\text{mix}}} = \frac{17275.67}{0.93518} = 18473 \text{ mL} = 18.473 \text{ L}$$

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20/20

$$\% \text{ error vol} = \frac{V_{\text{add}} - V_{\text{true}}}{V_{\text{true}}} \times 100\% = \frac{19.139 - 18.473}{18.473} \times 100\% = 3.60\%$$

$$\begin{aligned} \% \text{ error SG} &= \frac{SG_{\text{add}} - SG_{\text{true}}}{SG_{\text{true}}} \times 100\% = \frac{0.90264 - 0.93518}{0.93518} \times 100\% \\ &= -3.5\% < 0 \\ &\quad 3.5\% \text{ error SG low} \end{aligned}$$

- 3) $\left\{ \begin{array}{l} \text{By table 1.10, the glucose to adipic acid process loses } \$5400 \text{ per day.} \\ \text{By table 1.7, this process makes } \$27,100 \text{ /day} \end{array} \right.$

The cost for glucose need to drop: $\$ (27,100 - (-5,400)) = \$32,500$

But in table 1.10, the current cost is $\$48,500$

$\Rightarrow \$ (48,500 - 32,500) = \$16,000$ should be dropped

In table 1.10, it illustrated the consumption of glucose is $80,850 \text{ kg/day}$

$$\Rightarrow \text{Glucose price (new)} = \frac{\$16,000 / \text{day}}{80,850 \text{ kg/day}} = 0.198 \$ / \text{day}$$

$$\Rightarrow \text{The price drops needed} : 0.6 - 0.198 = 0.402 \$ / \text{kg}$$

- $\left\{ \begin{array}{l} \text{In table 1.11, the glucose to catechol process makes } \$49,200 / \text{day} \\ \text{In table 1.8, the benzene to catechol process makes } \$89,300 / \text{day} \end{array} \right.$
 $\hookrightarrow \Delta \$ / \text{day} = 40,100 \$ / \text{day}$

$$\Rightarrow \text{The 2nd glucose price} = \frac{\$40,100 / \text{day}}{80,850 \text{ kg/day}} = 0.496 \$ / \text{day}$$

$$\Rightarrow \text{The price should be} : 0.6 - 0.496 = 0.104 \$ / \text{kg}$$

20120